CSCE 40103: INTRO TO AI FOR SECURITY

LAB 3: MALWARE ANALYSIS USING CLUSTERING

Introduction

In this lab, you will use a pre-extracted PDF malware feature dataset to practice unsupervised clustering. The dataset description is available from the University of New Brunswick Canadian Institute for Cybersecurity: **CIC-Evasive-PDFMal2022 dataset description**. Use the provided CSV file on Blackboard for all lab work.

Dataset

Item	Requirement
Required file	*PDFMalware2022.csv*
Dataset source description	UNB CIC-Evasive-PDFMal2022 official dataset page
Data type	Static PDF file features in tabular form
Target or label column	`Class`, used only after clustering for interpretation
Main methods	K-Means, hierarchical clustering, DBSCAN
Evaluation	Silhouette score, cluster counts, noise points, cluster profiles, visual inspection
Submission file	`Lab3-LastName.ipynb`

Feature Groups

Group	Examples	Use in Lab
Identifiers and labels	FileName, Class, __index_level_0__	Remove from the feature matrix. Use Class only after clustering.
General PDF features	PdfSize, MetadataSize, Pages, TitleCharacters, isEncrypted, EmbeddedFiles, Images, Text, Header	Describe basic PDF properties.
PDF structure features	Obj, Endobj, Stream, Endstream, Xref, Trailer, StartXref, PageNo	Describe internal PDF structure.
Action and embedded-content features	JS, Javascript, AA, OpenAction, Acroform, JBIG2Decode, RichMedia, Launch, EmbeddedFile, XFA, Colors	Often useful for separating suspicious document behavior patterns.

Learning Objectives

- Load and inspect a PDF malware-feature dataset.
- Clean mixed numeric and object columns before clustering.
- Scale features before applying distance-based clustering methods.
- Apply K-Means and compare several K values using inertia and silhouette score.
- Build a dendrogram and compare linkage methods in hierarchical clustering.
- Apply DBSCAN and explain core points, border points, noise points, `eps`, and `MinPts`.
- Use silhouette score to compare clustering quality.
- Explain clusters using feature profiles and cautious interpretation.

Required Packages

Use these imports at the beginning of your notebook.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.metrics import silhouette_score
from sklearn.neighbors import NearestNeighbors
from scipy.cluster.hierarchy import linkage, dendrogram
```

Function and Method Parameters Used in This Lab

Function or Method	Important Parameters	Purpose
pd.read_csv()	filepath_or_buffer	Loads the CSV dataset.
df.info()	No required parameter	Shows column names, data types, and missing values.
pd.to_numeric()	errors="coerce"	Converts object columns to numeric when possible.
StandardScaler()	No required parameter	Scales numeric features before clustering.
PCA()	n_components=2, random_state=40103	Creates a 2D visualization view.
KMeans()	n_clusters, random_state, n_init	Runs K-Means clustering.
AgglomerativeClustering()	n_clusters, linkage	Runs hierarchical clustering.
linkage()	method	Creates the hierarchy used for a dendrogram.
DBSCAN()	eps, min_samples	Finds dense clusters and noise points.
silhouette_score()	X, labels	Measures cohesion and separation.

Instructions

- Create a Notebook named `Lab3-LastName.ipynb` on Google Colab.
- Place *PDFMalware2022.csv* in the same folder as your notebook.
- Use markdown headings for each lab step.
- Run every code cell before submitting.
- Include your name, date, course name, and lab number at the top.
- All explanations must be in your own words.

Important Cleaning Notes

- Drop `FileName`, `Class`, and `__index_level_0__` before clustering.
- Convert `Yes` and `No` values to `1` and `0` when they appear in feature columns.
- Convert numeric-looking object columns using `pd.to_numeric(..., errors="coerce")`.
- For `Header`, either extract the PDF version number or remove the column if conversion becomes messy.
- Fill missing numeric values after conversion, then scale the feature matrix.

Step 1: Create the Notebook and Load the Dataset

- Create `Lab3-LastName.ipynb`.
- Load the required packages.
- Load *PDFMalware2022.csv*.
- Display the first five rows and dataset shape.

Required outputs: First five rows and dataset shape.

Step 2: Inspect the Dataset

- Show column names and data types.
- Check missing values.
- Check duplicate rows.
- Display value counts for the `Class` column.
- Write 2 to 3 sentences explaining what each row represents and why `Class` should not be used during clustering.

Required outputs: Column summary, missing value summary, duplicate count, Class distribution, short explanation.

Step 3: Clean and Prepare Features

- Remove `FileName`, `Class`, and `__index_level_0__` from the feature matrix.
- Convert `Yes` and `No` values to `1` and `0` where needed.
- Convert object columns that contain numbers into numeric columns.
- Drop or transform `Header` clearly.
- Fill missing numeric values after conversion.
- Show the final selected feature list.

Required outputs: Cleaning summary, selected feature list, preview of cleaned feature matrix.

Step 4: Scale Features and Apply PCA

- Apply `StandardScaler` to the cleaned feature matrix.
- Create a 2D PCA view only for visualization.
- Plot the PCA view colored by `Class` for reference only.
- Explain why scaling is required for K-Means, hierarchical clustering, DBSCAN, and silhouette score.

Required outputs: Scaled feature preview, PCA plot, scaling explanation.

Step 5: K-Means Clustering

- Run K-Means for K values from 2 to 8.
- Record inertia and silhouette score for each K.
- Create an elbow plot.
- Create a silhouette score plot.
- Choose one K and explain why.
- Visualize the selected clustering result using a 2D PCA scatter plot.

Required outputs: Comparison table, elbow plot, silhouette plot, PCA cluster plot, short interpretation.

Step 6: Hierarchical Clustering and Linkage Methods

- Create a dendrogram.
- Run `AgglomerativeClustering` using the same K selected in Step 5, or justify a different K.
- Compare at least three linkage methods: `ward`, `complete`, `average`, and/or `single`.
- Create a 2D PCA scatter plot colored by hierarchical cluster.
- Explain how linkage changed the result.

Required outputs: Dendrogram, linkage comparison table, PCA plot, short linkage explanation.

Step 7: DBSCAN Clustering

- Create a k-distance plot to help choose `eps`.
- Run DBSCAN with at least three `eps` values.
- Try at least two `min_samples` values.
- Record number of clusters and number of noise points for each setting.
- Compute silhouette score only when valid.
- Create a 2D PCA scatter plot showing DBSCAN clusters and noise points.

Required outputs: K-distance plot, parameter comparison table, PCA plot, noise count, short explanation.

Step 8: Compare Methods and Interpret Clusters

- Create one comparison table for K-Means, hierarchical clustering, and DBSCAN.
- Include number of clusters, silhouette score when valid, number of noise points if applicable, and short notes.
- Create cluster profiles using mean feature values grouped by cluster.
- Compare clusters against the `Class` column only after clustering.
- Explain which method produced the most useful result for this dataset.

Required outputs: Method comparison table, cluster profile table, Class comparison table, 4 to 6 sentence interpretation.

Required Method Comparison Table

Method	Parameters Used	Number of Clusters	Noise Points	Silhouette Score	Short Interpretation
K-Means	K = ?	?	N/A	?	
Hierarchical	K = ?, linkage = ?	?	N/A	?	
DBSCAN	eps = ?, min_samples = ?	?	?	?	

Step 9: Final Review

- Notebook must be organized, readable, and fully executed.
- Add comments or provide explanation of each step.
- Plots and tables must have meaningful titles.

Required output: Clean final notebook with executed cells, titled plots, and readable explanations.

Reference Guides

Linkage Method Guide

Linkage	Plain Meaning	Common Behavior
Single	Distance between the closest points in two clusters	Can create chain-like clusters.
Complete	Distance between the farthest points in two clusters	Tends to create tighter clusters.
Average	Average distance between points across two clusters	Often gives a balanced result.
Ward	Merges clusters while minimizing added within-cluster spread	Works well for compact numeric clusters.

DBSCAN Parameter Guide

Parameter or Output	Meaning	What to Watch
`eps`	Neighborhood radius around each point	Too small may label many points as noise. Too large may merge groups.
`min_samples`	Minimum points needed to form a dense region	Higher values require stronger density.
Core point	Has enough neighbors within `eps`	Starts or expands a cluster.
Border point	Near a core point but not dense enough by itself	Belongs to a cluster but does not expand it.
Noise point	Not reachable from a core point	May be unusual or may reflect poor parameter choice.

What to Submit

- Notebook filename is `Lab3-LastName.ipynb`.
- *PDFMalware2022.csv* loads correctly.
- All code cells are executed.
- Dataset cleaning decisions are explained.
- K-Means, hierarchical clustering, and DBSCAN are included.
- Silhouette score is used correctly.
- Cluster profiles and interpretations are included.
- Notebook is clean, organized, and fully executed.

Grading Rubric

Part	Points
Step 1: Create notebook and load dataset	5
Step 2: Inspect dataset	10
Step 3: Clean and prepare features	15
Step 4: Scale features and Apply PCA	10
Step 5: K-Means clustering	15
Step 6: Hierarchical clustering and linkage methods	15
Step 7: DBSCAN clustering	15
Step 8: Compare methods and interpret clusters	10
Step 9: Comments/Explanations	5
Total	**100**